

**NATIONAL JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS**
Higher 2

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

Paper 1 Multiple Choice

9746/01

Tues 22 September 2009

1 hour

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, subject class and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

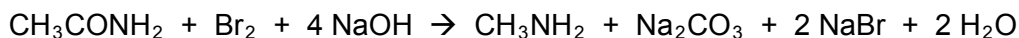
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 The Hofmann degradation is an example of a step-down reaction between an amide and a mixture of bromine and sodium hydroxide solution to form amines.

The following shows an equation for the preparation of methylamine:



What is the mass of sodium hydroxide required for the reaction, given that 16.5 g of methylamine is produced?

[A_r of C = 12.0; H = 1.0; N = 14.0; O = 16.0; Na = 23.0]

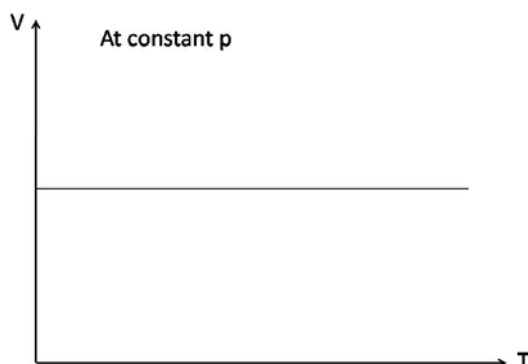
- A 5.32 g
B 21.3 g
C 66.0 g
D 85.2 g
- 2 An element X forms an ion X^{n+} . The atom X and the ion X^{n+} have the same
- A size
B electronic configuration
C number of protons
D number of electrons
- 3 *Use of the Data Booklet is relevant to this question.*

Which of the following is the electronic configuration of iron in $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{4-}$?

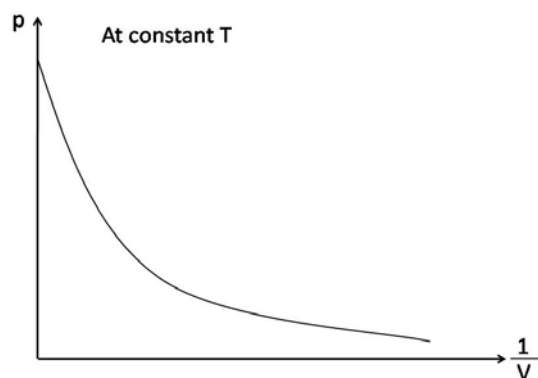
- A $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
B $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
C $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$

- 4 Which graph correctly describes the behavior of a fixed mass of an ideal gas?
(T is measured in K.)

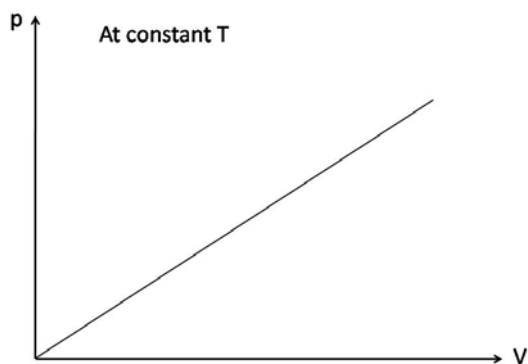
A



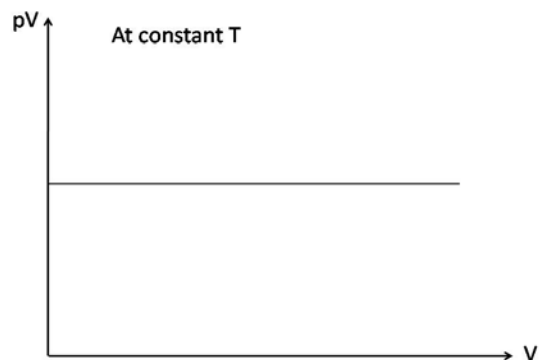
C



B



D



- 5 A sample of an organic compound of molar mass M is vapourised in a gas syringe and occupies $V \text{ cm}^3$ at $T \text{ K}$ and $p \text{ atm}$.

What is the density of the organic compound?

A $\rho = \frac{p \times M \times (T + 273)}{22400 \times 273}$

B $\rho = \frac{p \times M \times 273}{22400 \times T}$

C $\rho = \frac{p \times M}{22400 \times 273 \times T}$

D $\rho = \frac{M \times (T + 273)}{p \times 22400 \times 273}$

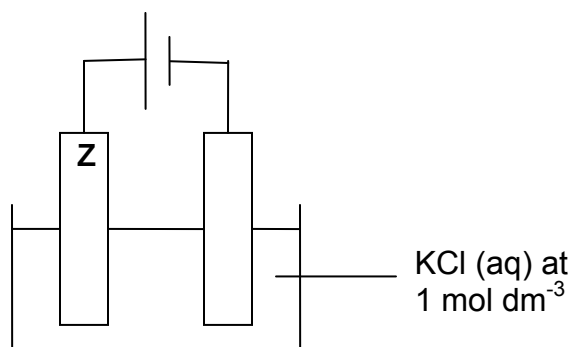
- 6 0.194 g of chromium is deposited on the cathode during electrolysis under suitable conditions when a constant current of 0.5 A is passed into a chromium-containing electrolyte for 36 minutes.

Which of the following could have been the electrolyte?

[A_r of Cr = 52.0]

- A CrCl_2
- B CrCl_3
- C $\text{K}_2\text{Cr}_2\text{O}_3$
- D $\text{K}_2\text{Cr}_2\text{O}_7$

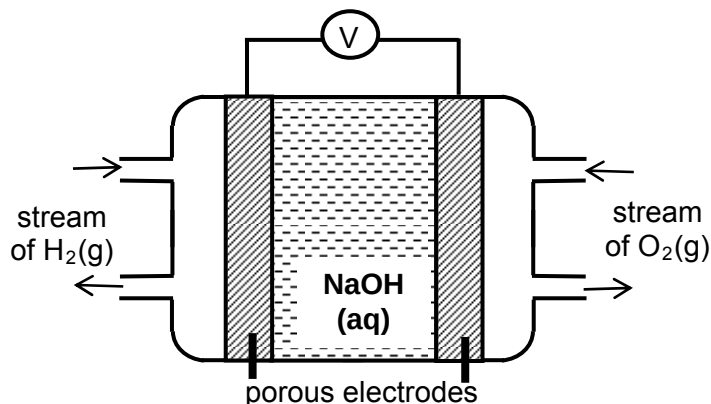
- 7 The diagram below is an experimental set-up to show the corrosion of iron. Both electrodes are made of iron.



What will happen at electrode Z when a large current is passed through KCl solution?

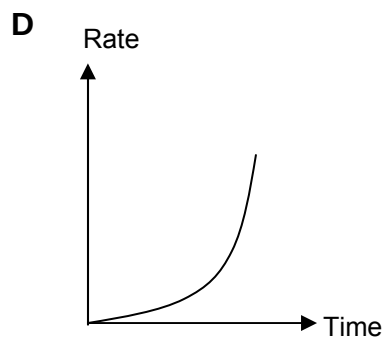
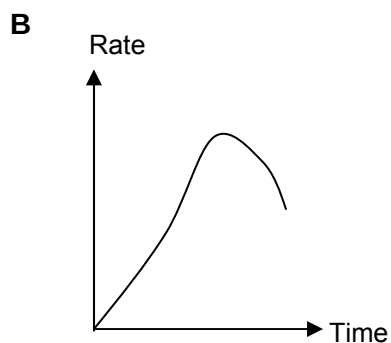
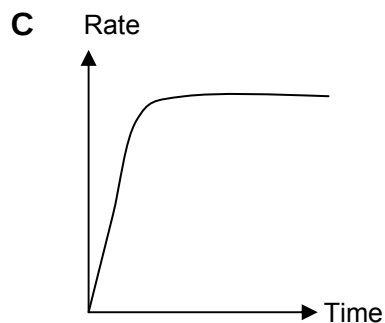
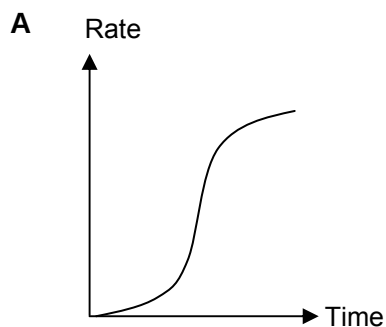
- A Corrosion of Fe and O_2 gas evolved
- B Corrosion of Fe and Cl_2 gas evolved
- C No corrosion of Fe and H_2 gas evolved
- D No corrosion of Fe and K deposited

- 8 A hydrogen-oxygen fuel cell is built using 1.00 mol dm^{-3} sodium hydroxide as the electrolyte. What would be the change in pH of the solution around each electrode when current flows?

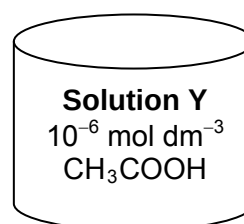
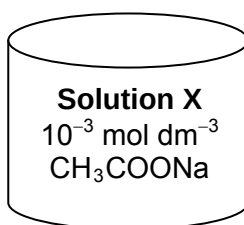
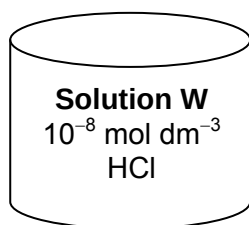


- | | Anode | Cathode |
|----------|--------------|----------------|
| A | decrease | increase |
| B | decrease | decrease |
| C | increase | decrease |
| D | increase | increase |
- 9 Which is the correct statement about the following reaction?
- $$\text{P(g)} + \text{Q(g)} \xrightleftharpoons{\text{M(s)}} \text{R(g)} \quad \Delta H < 0$$
- A** Both the rate constant and equilibrium constant will increase with increasing temperature.
- B** The solid **M** will lower the activation energy of both forward and backward reactions.
- C** Increasing temperature will lower the activation energy resulting only in a greater fraction of **R** molecules with energy greater than activation energy.
- D** The activation energy of the forward reaction is equal to the activation energy of the backward reaction.

- 10 Which graph correctly represents an auto-catalytic reaction (reaction in which one of the products catalyses the reaction)?



- 11 Which sequence represents the solutions in increasing order of their pH?
(Given: K_a of ethanoic acid is 1.75×10^{-5})

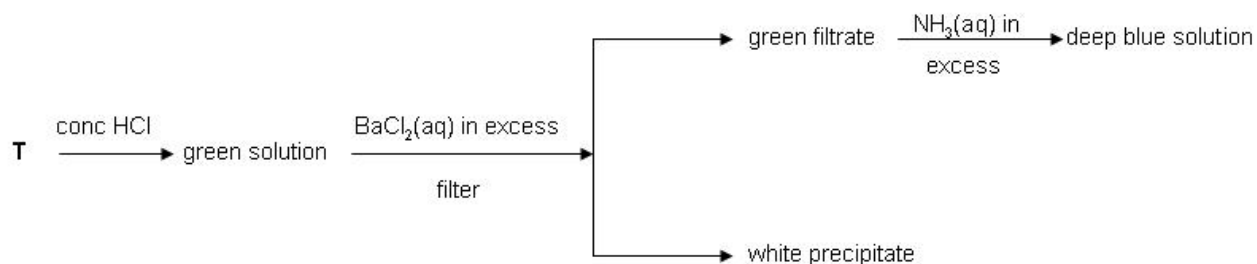


- A W, X, Y
B Y, W, X
C W, Y, X
D Y, X, W
- 12 What is the solubility of $\text{Mg}(\text{OH})_2$ in 0.1 mol dm^{-3} aqueous NH_3 ?
(Given: K_{sp} of $\text{Mg}(\text{OH})_2$ is 1.80×10^{-10} ; K_b of aqueous NH_3 is 1.75×10^{-5})

- A $1.80 \times 10^{-8} \text{ mol dm}^{-3}$
B $2.57 \times 10^{-5} \text{ mol dm}^{-3}$
C $1.03 \times 10^{-4} \text{ mol dm}^{-3}$
D $3.56 \times 10^{-4} \text{ mol dm}^{-3}$

- 13** Which of the following solutions will give a precipitate when mixed?
- A** $\text{Ba}(\text{NO}_3)_2$ and NH_3
 - B** $\text{Mg}(\text{NO}_3)_2$ and H_2SO_4
 - C** $\text{Ba}(\text{NO}_3)_2$ and HCl
 - D** $\text{Sr}(\text{NO}_3)_2$ and H_2SO_4
- 14** Sodium chlorate(V), NaClO_3 , is obtained by electrolyzing
- A** a dilute solution of NaCl at room temperature.
 - B** a dilute solution of NaCl at 70°C .
 - C** a concentrated solution of NaCl at room temperature.
 - D** a concentrated solution of NaCl at 70°C .
- 15** The reaction of 0.01 mole of chromium (III) chloride $[\text{Cr}(\text{H}_2\text{O})_{6-x}\text{Cl}_x]\text{Cl}_y$ with excess AgNO_3 results in the precipitation of 2.87g of AgCl . Which one of the following regarding the chromium complex is correct?
[A_r of Ag : 108; Cl : 35.5]
- | | x | y | Shape of complex |
|----------|----------|----------|-------------------------|
| A | 2 | 1 | trigonal bipyramidal |
| B | 2 | 1 | octahedral |
| C | 1 | 2 | trigonal bipyramidal |
| D | 1 | 2 | octahedral |
- 16** The oxide of a metal **X** is insoluble in water but it dissolves in aqueous ammonia to form a complex cation $[\text{X}(\text{NH}_3)_2]^+$. On reacting with dilute sulphuric acid, the metal oxide gives a blue solution and a deposit of the metal **X**. What is the formula of the metal oxide?
- A** CuO
 - B** Cu_2O
 - C** Ag_2O
 - D** Fe_2O_3

17 Below is a scheme of reaction for a salt T.



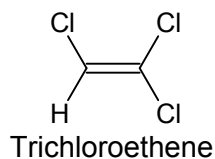
What is T?

- A chromium (III) chloride
- B chromium (III) sulphate
- C copper (II) chloride
- D copper (II) sulphate

18 How many isomeric carbonyl compounds with molecular formula $C_5H_{10}O$ are there?

- A 7
- B 8
- C 9
- D 10

19 Trichloroethene is widely used as dry-cleaning agent.



Which of the following does trichloroethene react with to give a chiral product?

- A Br_2
- B HCl
- C $NaOH$
- D $NaCN$

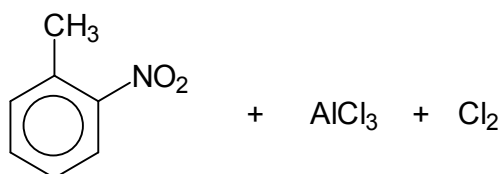
- 20 The oxo-reaction is an important industrial process in which an alkene combines directly with carbon monoxide and hydrogen under suitable conditions. The reaction with ethene is shown below.



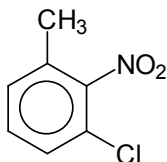
Which of the following structural formula correctly represents the product of the oxo-reaction starting with but-2-ene ?

- A $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHO}$
 B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
 C $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
 D $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$

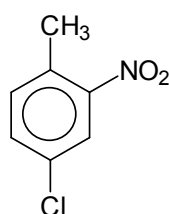
- 21 Which is the major product for the following reaction?



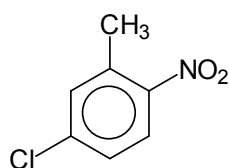
A



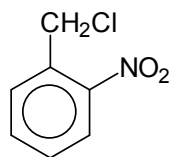
C



B



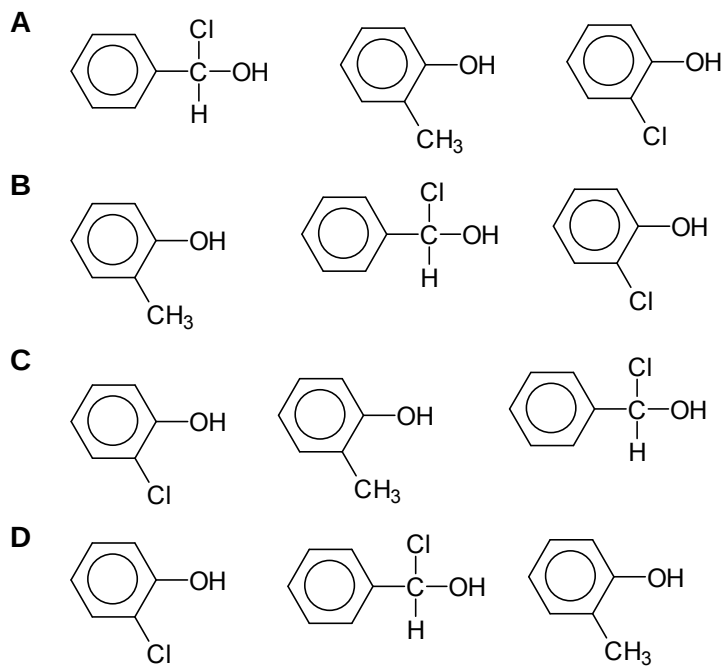
D



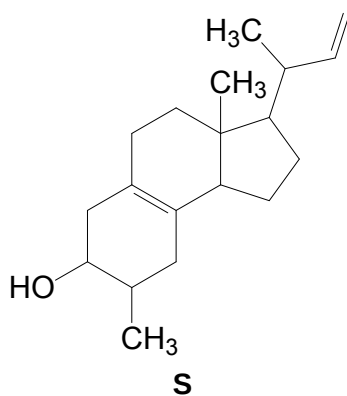
- 22 Which of the following will **not** be produced when 1-chloropropane is heated in ethanolic sodium hydroxide?

- A $\text{CH}_3\text{CH}=\text{CH}_2$
 B $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{ONa}$
 D $\text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_3$

23 Which sequence correctly represents the compounds in the order of increasing pK_a values?

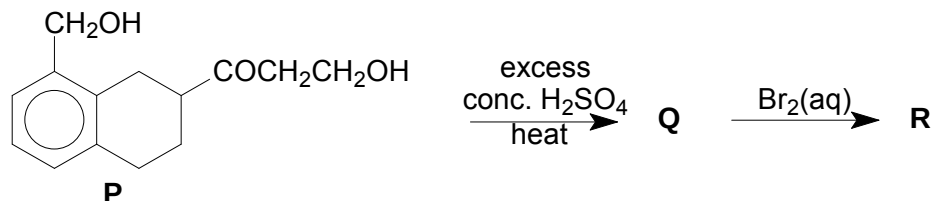


24 How many chiral centres are present in the **major** product formed when compound **S** undergoes a reaction with KBr and concentrated H_2SO_4 at high temperature?

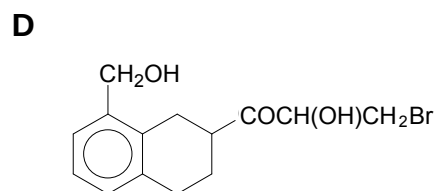
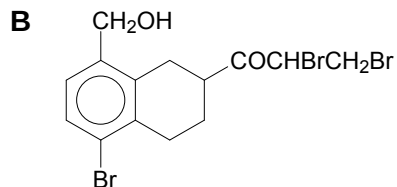
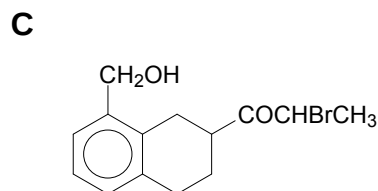
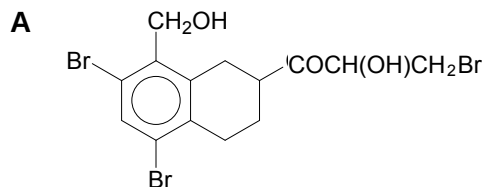


- A** 6
B 7
C 8
D 9

25 Compound **P** is used in the following synthesis.



What is the likely structure of the final product **R**?



26 Compound **N** is a product formed from the reaction between an alcohol and hot acidified potassium manganate(VII).

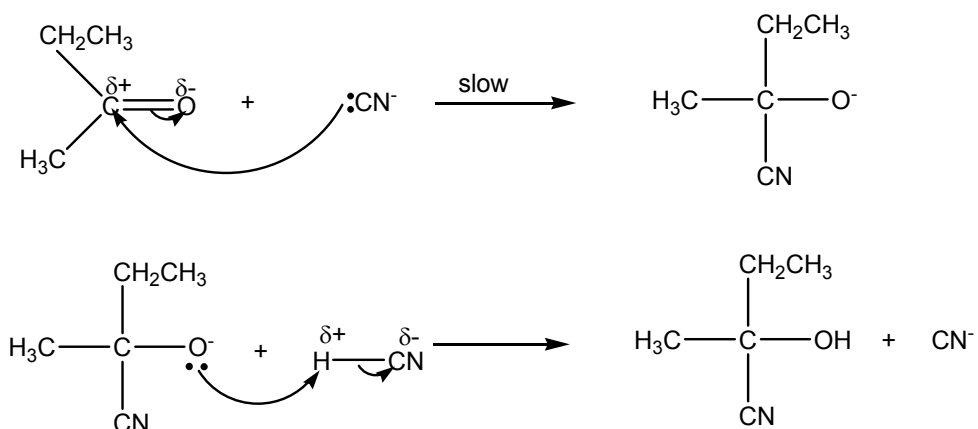
Compound **N**

- gives no yellow precipitate with alkaline iodine
- evolves a colourless gas when treated with aqueous sodium carbonate
- reacts with 2 moles of PCl_5 to give white fumes

What is the structural formula of the alcohol?

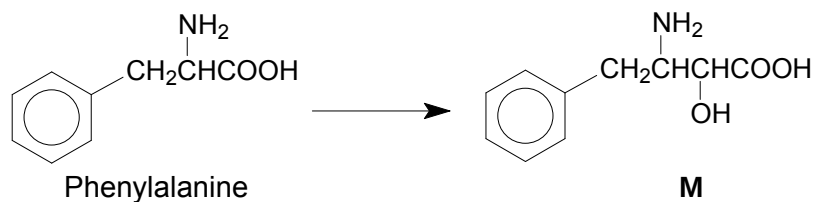
- A** $\text{CH}_3\text{COCH}_2\text{CH}_2\text{COOH}$
- B** $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- C** $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{COOH}$
- D** $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CH}_2\text{OH}$

27 The mechanism below shows how butanone reacts with hydrogen cyanide.



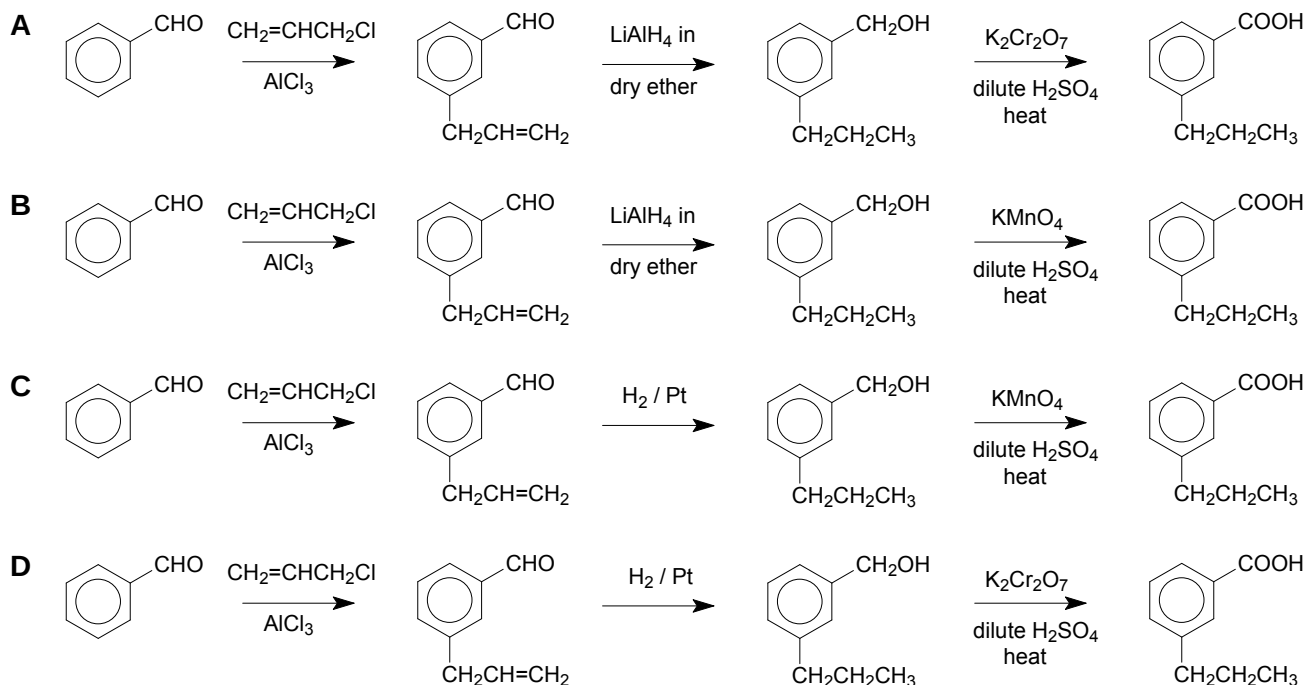
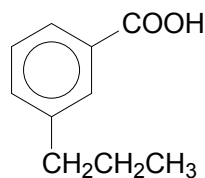
Which statement is true?

- A The product mixture is optically active.
 - B Rate of reaction increases when butanone is substituted with butanal.
 - C Cyanohydrin produced can react with ammonia at high pressure to form amine.
 - D Cyanohydrin produced requires potassium manganate(VII) to form carboxylic acid .
- 28 Phenylalanine is used to synthesize compound **M** during the production of a drug. Which of the following can be used to identify the phenylalanine from compound **M**?



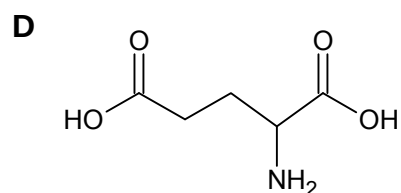
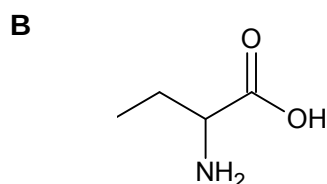
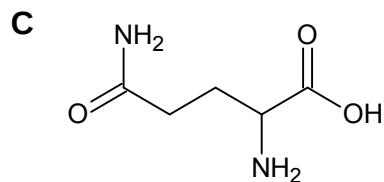
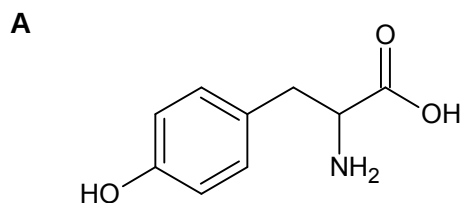
- A Sodium metal
- B Phosphorus pentachloride
- C Aqueous sodium carbonate
- D Hot acidified potassium dichromate(VI)

29 Which synthetic route will lead to a successful synthesis of the following product?



30 Electrophoresis is a technique of separating and identifying amino acids. A solution of amino acids is absorbed into paper that is moistened with a buffer solution and stretched between two electrodes. Positively charged species move towards the cathode, negatively charged species towards the anode.

With a buffer at pH 4, which amino acid will move most readily towards the cathode?



Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which statements about a 12.0 g sample of ^{12}C are correct?

- 1** The number of atoms in the sample is 6.02×10^{23} .
- 2** The number of atoms in the sample is the same as the number of atoms in 4.0 g of helium.
- 3** The number of atoms in the sample is the same as the number of atoms in 28.0 g of nitrogen gas.

32 *Use of the Data Booklet is relevant to this question.*

An ion X^{2-} has 36 electrons and 45 neutrons. Which of the following statements are true?

- 1** It is isoelectronic with Rb^+ .
- 2** It has a larger radius than Sr^{2+} .
- 3** More energy is required to remove an electron from X^- than from X^{2-} .

33 In which pair is the bond angle in the first species greater than that in the second species?

- 1** H_2O , H_2S
- 2** CO_3^{2-} , BrF_3
- 3** CH_4 , CCl_4

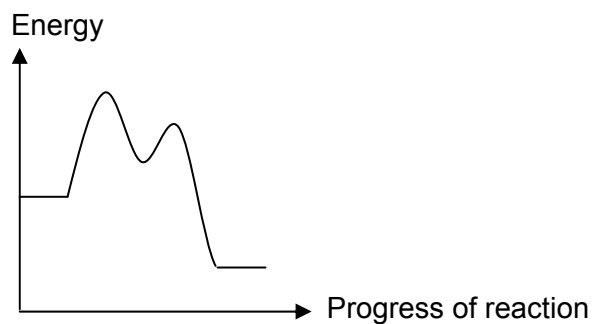
- 34 Which of the following statements can be deduced from the following standard potentials listed below?

	E^θ / V
$\text{Cd}^{2+}(\text{aq}) + 2\text{e} \rightleftharpoons \text{Cd}(\text{s})$	-0.40
$\text{Co}^{2+}(\text{aq}) + 2\text{e} \rightleftharpoons \text{Co}(\text{s})$	-0.28
$\text{Co}^{3+}(\text{aq}) + \text{e} \rightleftharpoons \text{Co}^{2+}(\text{aq})$	$+1.82$
$\text{Ga}^{3+}(\text{aq}) + 3\text{e} \rightleftharpoons \text{Ga}(\text{s})$	-0.56
$\text{In}^{3+}(\text{aq}) + 3\text{e} \rightleftharpoons \text{In}(\text{s})$	-0.34

- 1 Co^{2+} is a stronger oxidizing agent than Ga^{3+} .
 - 2 In is a weaker reducing agent than Cd.
 - 3 Co^{3+} can be reduced to Co by In.
- 35 In a chemical reaction, J reacts with K to form L. The initial rates of the reaction are measured in the following experiment.

Experiment	$[\text{J}] / \text{mol dm}^{-3}$	$[\text{K}] / \text{mol dm}^{-3}$	Initial rate / mol dm^{-3}
1	0.150	0.250	2.80×10^{-5}
2	0.150	0.500	5.60×10^{-5}
3	0.075	0.500	2.80×10^{-5}
4	0.075	0.250	1.40×10^{-5}

The energy profile diagram for the reaction is as shown:



Which of the following are possible overall equations of the above reaction?

- 1 $\text{J} + \text{K} \rightarrow \text{L}$
- 2 $2\text{J} + \text{K} \rightarrow \text{L}$
- 3 $\text{J} + 2\text{K} \rightarrow \text{L}$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

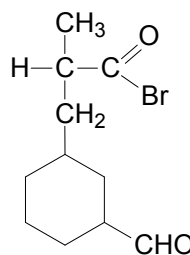
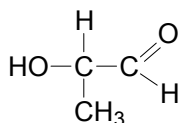
36 Which of the following properties will give a decreasing trend down Group VII?

- 1 Thermal stability of hydrogen halides
- 2 Reducing power of halides
- 3 Boiling point of halogens

37 Samples of the gases CH_3Cl and Cl_2 are mixed together and irradiated with light. Which statements are true?

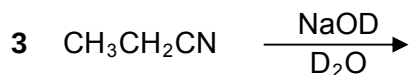
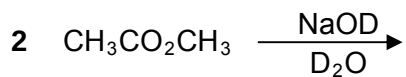
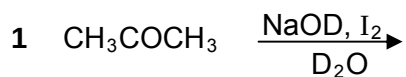
- 1 Chlorine atoms are formed in the initiation step.
- 2 The chlorine radical acts as a catalyst in the propagation steps.
- 3 Ethane is produced in trace amounts in the termination step.

38 Which reagents can be used to distinguish between these two compounds?

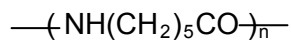


- 1 Fehling's solution
- 2 Silver nitrate solution
- 3 Alkaline iodine solution

- 39 Which reactions yield a carbon compound incorporating deuterium, D?
[D = ^2H]



- 40 Nylon is a biodegradable polymer used in clothes manufacturing.
A section of the nylon molecule shown below:



Which statement about nylon is true?

- 1 It contains amide linkages.
- 2 It is basic.
- 3 It gives an orange precipitate with 2,4-dinitrophenylhydrazine.